

An analysis of China's grain production: looking back and looking forward : Li et. Al. (2013)



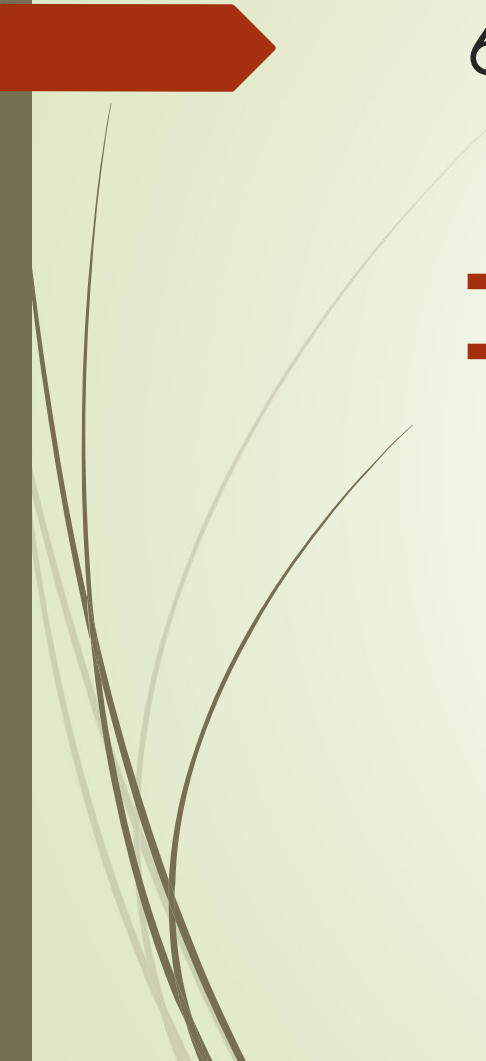
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Goal of Paper

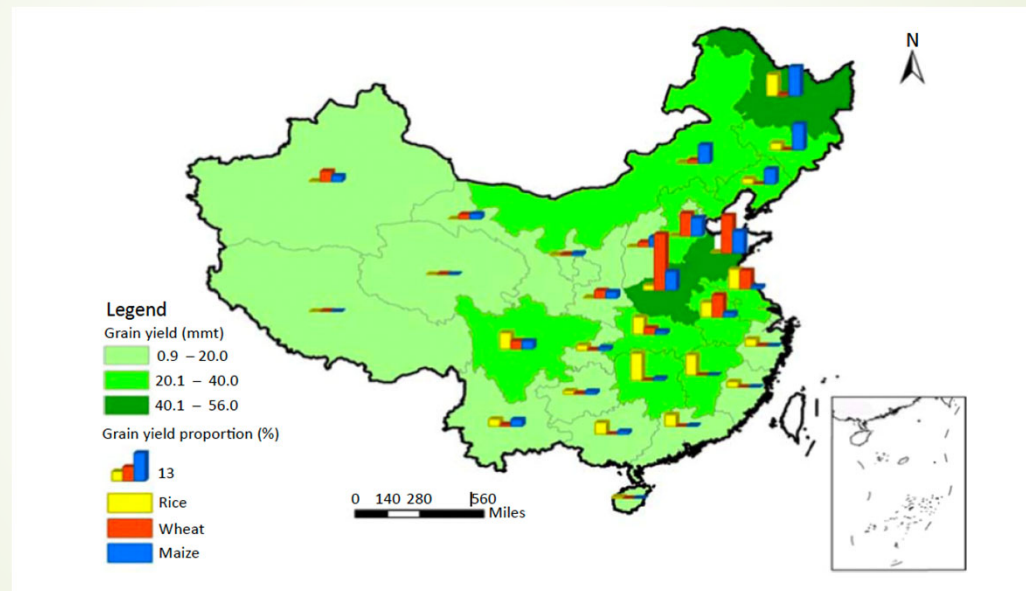
- Published in “Food and Energy Security” – 9 authors
- The big question : how can China sustainably grow its agricultural output to meet increasing internal demand in the coming decades?
 - Analysis of 60 years of trends in grain production, relevant policies and influencing factors
 - Project China's grain demand for 2030 and major production constraints
 - Identify pathways for sustainable development of grain production in the future



60 years of grain production

- Total grain output = cultivated area + crop yield
- General influencing factors: weather conditions, technological advancement, government policies
 - 1953 – 1978 : state-controlled commune system
 - 1978 – 1984 : land use reform + improvement in technology
 - 1984 – 1998 : yield increase through technology, marketization
 - 1998 : Asian financial crisis – development of cash crops
 - 2004 – 2013 : changes in agricultural subsidies

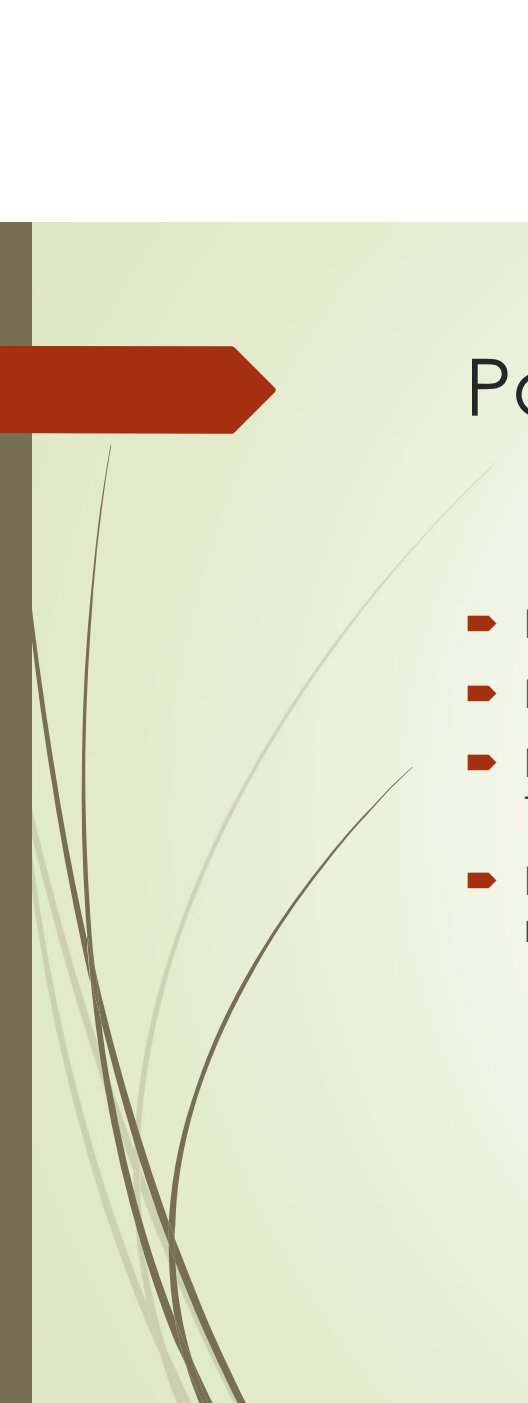
3 Main Areas of Grain Production





Projection for Grain Demand in 2030

- China produced 571 MT of grain in 2011, will need 776 MT by 2030
- Demand growth = urbanization + rise in income + population growth
- Assuming crop yield remains the same, cultivated area would need to increase by 36.4% by 2030
 - Crop yield is contingent on genetic and crop improvement programs
 - Area Constraints : soil erosion, water shortage, land competition



Pathways for Sustainable Development

- How to improve farmers' initiatives for grain production? *Subsidy reform*
- How to further boost crop yield? *Technology & innovation, infrastructure*
- How to eliminate/minimize the environmental costs of grain production? *Technology and innovation*
- How to recouple grain production with animal production to better recycle manure and protect water quality? *Technology and innovation*



Paper Reflection

Strength

- Good basic overview
- Strong literature review
- Considered multiple factors

Weakness

- Poor organization
- Non-linear
- Overly-ambitious – what was the main goal or contribution?
- Did not clear identify policy pathways